

Introduction to the Root-Locus Method

Where the closed-loop poles go as the loop gain varies from 0 to infinity.

Ogata, *Modern Control Engineering*, Ch. 6.

In this tutorial you will:

- read the **angle** and **magnitude** conditions that define the locus,
- learn the construction rules (branches, asymptotes, breakaway points), and
- draw and read a locus with `rlocus`.

Step through with **Ctrl+Enter**, or render a report with **publish**.

The root locus is the set of all locations in the s -plane that the closed-loop poles can occupy as a single real parameter (almost always the loop gain K) is varied from 0 to ∞ . For a unity-feedback system with open-loop transfer function $KG(s)$, the closed-loop characteristic equation is

$$1 + KG(s) = 0 \quad \Leftrightarrow \quad G(s) = -\frac{1}{K}$$

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The Angle and Magnitude Conditions

Writing $G(s)$ in pole-zero form,

$$G(s) = \frac{\prod_{j=1}^m (s - z_j)}{\prod_{i=1}^n (s - p_i)}$$

a point s_0 lies on the root locus for some $K > 0$ if and only if it satisfies the **angle condition**

$$\sum_{j=1}^m \angle(s_0 - z_j) - \sum_{i=1}^n \angle(s_0 - p_i) = \pm 180^\circ(2k + 1), \quad k = 0, 1, 2, \dots$$

The corresponding gain at that point is then found from the **magnitude condition**

$$K = \frac{\prod_{i=1}^n |s_0 - p_i|}{\prod_{j=1}^m |s_0 - z_j|}$$

The angle condition alone determines the **shape** of the locus (independent of K); the magnitude condition is then used to read off the gain at any particular point on that shape.

Basic Construction Rules (Summary)

For $KG(s)H(s)$ with n poles and m zeros ($n \geq m$):

1. The locus has n branches, starting ($K = 0$) at the open-loop poles and ending ($K = \infty$) at the open-loop zeros or at infinity.
2. The locus is symmetric about the real axis.
3. A point on the real axis is part of the locus if the total number of real poles and zeros to its right is odd.
4. $n - m$ branches go to infinity along asymptotes centered at $\sigma_a = \frac{\sum p_i - \sum z_j}{n - m}$ with angles $\theta_a = \frac{180^\circ(2k+1)}{n - m}$.
5. Breakaway/break-in points on the real axis occur where $\frac{dK}{ds} = 0$.

Worked Verification of the Angle Condition

Consider $G(s) = \frac{1}{s(s+2)(s+4)}$ and test whether $s_0 = -1 + j1.5$ lies (approximately) on the locus.

```
poles = [0 -2 -4];
s0 = -1 + 1.5j;

angles = angle(s0 - poles) * 180/pi;
total_angle = sum(angles);
fprintf('Pole angles at s0: %s degrees\n', mat2str(angles,4))
fprintf('Sum of pole angles = %.2f degrees\n', total_angle)

% With no finite zeros, the open-loop phase is angle(G(s0)) = -sum(pole
% angles). The angle condition is satisfied (s0 lies on the locus) when
% this phase is an odd multiple of 180 deg, i.e. when the residual below
% is zero.
ol_phase = -total_angle;
resid = mod(ol_phase, 360) - 180;    % 0 == exactly on the locus
fprintf('Open-loop phase at s0 = %.2f deg\n', ol_phase)
fprintf('Deviation from the angle condition = %.2f deg\n', resid)
if abs(resid) < 1
    fprintf('s0 satisfies the angle condition (it lies on the locus).\n')
else
    fprintf('s0 is off the locus by %.2f deg (this test point is only approximate).\n', resid)
end

% Find the gain that would place a closed-loop pole at s0 (magnitude
% condition); only exactly valid once s0 is on the locus.
K_at_s0 = abs(prod(s0-poles));
fprintf('Magnitude-condition gain at s0: K = %.4f\n', K_at_s0)

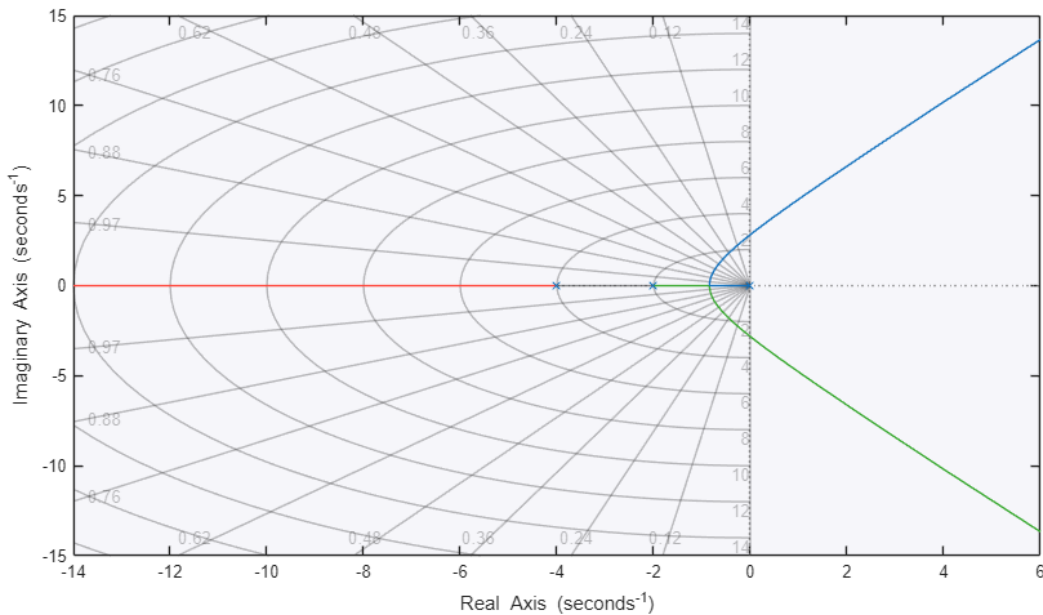
Pole angles at s0: [123.7 56.31 26.57] degrees
Sum of pole angles = 206.57 degrees
Open-loop phase at s0 = -206.57 deg
Deviation from the angle condition = -26.57 deg
s0 is off the locus by -26.57 deg (this test point is only approximate).
Magnitude-condition gain at s0: K = 10.9008
```

Plotting the Locus Directly

In practice, `rlocus` automates the angle/magnitude bookkeeping and sweeps K over a default (or specified) range.

```
G = tf(1,poly(poles));
figure
rlocus(G)
title('Root Locus of $G(s)=\frac{1}{s(s+2)(s+4)}$', 'Interpreter', 'latex', 'FontSize', 20)
grid on
```

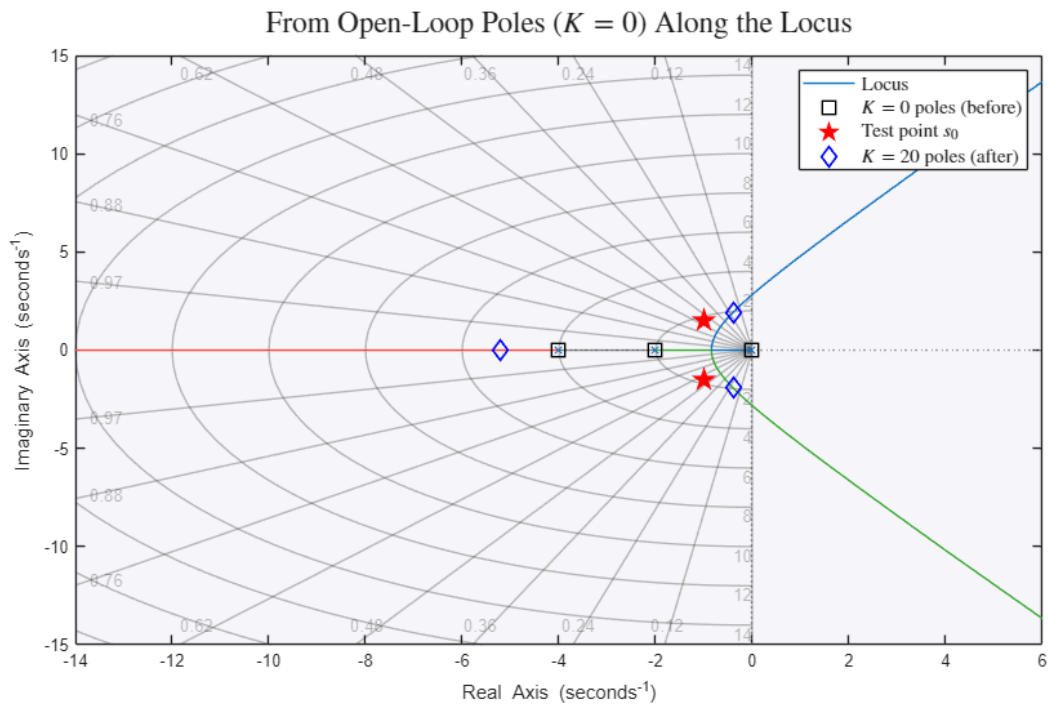
$$\text{Root Locus of } G(s) = \frac{1}{s(s+2)(s+4)}$$



Before vs. After: Starting Points and Where They Go

At $K=0$ the closed-loop poles coincide with the open-loop poles (the "before"); as K increases they trace the locus. Overlay the $K=0$ poles, the test point s_0 , and the closed-loop poles for a sample K so the migration the locus encodes is explicit.

```
K_demo = 20;
cl_demo = pole(feedback(K_demo*G,1));
figure
rlocus(G)
hold on
plot(real(poles),imag(poles),'ks','MarkerSize',10,'LineWidth',1.5)
plot([real(s0) real(s0)], [imag(s0) -imag(s0)], 'rp','MarkerSize',13,'MarkerFaceColor','r')
plot(real(cl_demo),imag(cl_demo),'bd','MarkerSize',8,'LineWidth',1.5)
hold off
legend('Locus','$K=0$ poles (before)','Test point $s_0$', '$K=20$ poles (after)', ...
      'Interpreter','latex','FontSize',11)
title('From Open-Loop Poles ($K=0$) Along the Locus','Interpreter','latex','FontSize',17)
grid on % keep rlocus's own axis labels (it re-appends units, which breaks
        % a custom latex xlabel/ylabel)
```



Try it yourself

- Move the test point to $s_0 = -1 + 2j$ and see whether the angle- condition residual gets closer to zero (nearer the true locus).
- Add a zero ($G = \text{tf}([1 \ 3], \text{poly}(\text{poles}))$) and notice the branches bend toward it instead of all running to infinity.